

SECURITY OPERATIONS CENTER

Security Incident Investigation Report



Multi-Stage Attack Analysis using Wazuh SIEM

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April 23, 2026

SECURITY INCIDENT INVESTIGATION REPORT

Multi-Stage Attack Analysis using Wazuh SIEM

1. EXECUTIVE SUMMARY

- A multi-stage attack simulation was conducted on endpoint **DESKTOP-KA2LJHB** within the LAB domain, successfully replicating real-world adversary behavior across six distinct attack phases. The attack chain was detected and correlated by Wazuh SIEM, generating six individual alerts that when analyzed in sequence, it reveal a complete compromise lifecycle.
- The investigation is based on event correlation, behavioral analysis, and timeline reconstruction. Each alert was individually triaged, contextually analyzed, and correlated into a unified attack chain demonstrating how a threat actor would progress from initial credential access through to evidence destruction.



SECURITY INCIDENT INVESTIGATION REPORT

Multi-Stage Attack Analysis using Wazuh SIEM

2. ENVIRONMENT OVERVIEW

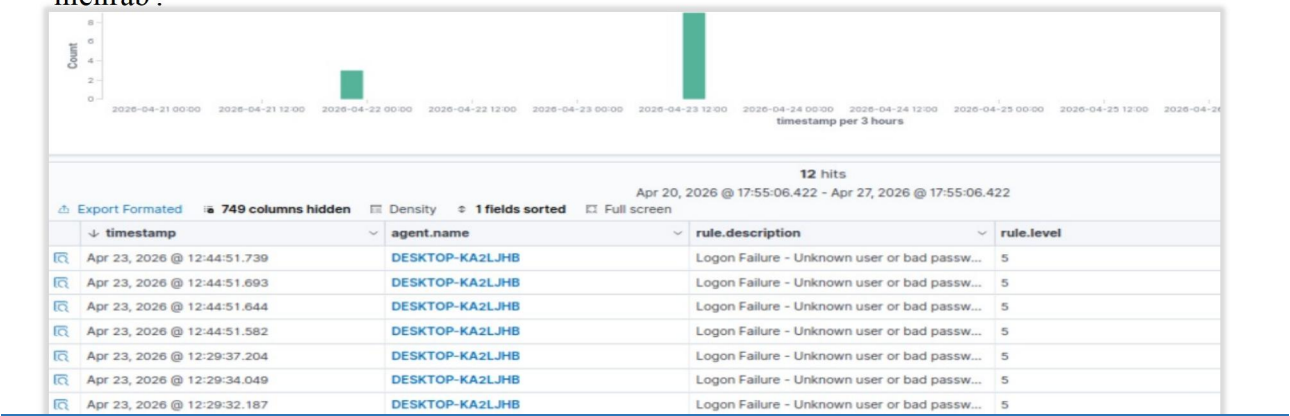
Endpoint Hostname	DSKTOP-KA2LJHB
Endpoint- IP Address	192.168.X.X
Domain	LAB
SIEM Platform	Wazuh SIEM v4.x
Log Sources	Windows Security Logs, Sysmon, Windows Defender AV
Agent ID	005
SIEM Manager	Ubuntu (Wazuh-Manager)
Analyst	Mehrab Shaheen(SOC Analyst)
Investigation Date	April 23, 2026

3. ALERT-BY-ALERT INVESTIGATION

Each of the following six alerts was individually analyzed using the 5W methodology (Who, What, When, Where, Why), followed by technical detail extraction, behavioral analysis, and final verdict determination.

Alert 1 — Failed Logon | Credential Access | Event ID: 4625

- To simulate an initial access attempt, multiple failed login attempts were performed on the target system using incorrect credentials for the account 'mehrab'.



SECURITY INCIDENT INVESTIGATION REPORT

Multi-Stage Attack Analysis using Wazuh SIEM

Document Details		View surrounding documents
agent.ip		
agent.name	DESKTOP-KA2LJHB	
data.win.eventdata.authenticationPackageName	Negotiate	
data.win.eventdata.failureReason	%2313	
data.win.eventdata.ipAddress	127.0.0.1	
data.win.eventdata.ipPort	0	
data.win.eventdata.keyLength	0	
data.win.eventdata.logonProcessName	User32	
data.win.eventdata.logonType	2	
data.win.eventdata.processId	0x1c0	
data.win.eventdata.processName	C:\Windows\System32\svchost.exe	
data.win.eventdata.status	0xc000006d	
data.win.eventdata.subStatus	0xc000006a	
data.win.eventdata.subjectDomainName	LAB	
data.win.eventdata.subjectLogonId	0x3e7	
data.win.eventdata.subjectUserName	DESKTOP-KA2LJHB\$	
data.win.eventdata.subjectUserSid	S-1-5-18	
data.win.eventdata.targetDomainName	DESKTOP-KA2LJHB	
data.win.eventdata.targetUserName	mehrab	
data.win.eventdata.targetUserSid	S-1-0-0	
data.win.eventdata.workstationName	DESKTOP-KA2LJHB	
data.win.system.channel	Security	
data.win.system.computer	DESKTOP-KA2LJHB.lab.local	
data.win.system.eventID	4625	

me	
data.win.system.channel	Security
data.win.system.computer	DESKTOP-KA2LJHB.lab.local
data.win.system.eventID	4625
data.win.system.eventRecordID	92510
data.win.system.keywords	0x8010000000000000
data.win.system.level	0
data.win.system.message	'An account failed to log on. Subject: Security ID: S-1-5-18 Account Name: DESKTOP-KA2LJHB \$ Account Domain: LAB Logon ID: 0x3e7 Logon Type: 2 Account For Which Logon Failed: Security ID: S-1-0-0 Account Name: mehrab Account Domain: DESKTOP-KA2LJHB Failure Information: Failure Reason: Unknown user name or bad password. Status: 0xc000006d Sub Status: 0xc000006a Process Information: Caller Process ID: 0x1c0 Caller Process Name: C:\Windows\System32\svchost.exe Network Information: Workstation Name: DESKTOP-KA2LJHB Source Network Address: 127.0.0.1 Source Port: 0 Detailed Authentication Information: Logon Process: User32
data.win.system.opcode	0
data.win.system.processID	668
data.win.system.providerGuid	{54849625-5478-4994-a5ba-3e3b0328c30d}
data.win.system.providerName	Microsoft-Windows-Security-Auditing
data.win.system.severityValue	AUDIT_FAILURE
data.win.system.systemTime	2026-04-23T07:44:17.7970486Z
data.win.system.task	12544
data.win.system.threadID	716
data.win.system.version	0
decoder.name	windows_eventchannel
id	1776930291.22638753
input.type	log
location	EventChannel
manager.name	mehrab-virtual-machine
rule.description	Logon Failure - Unknown user or bad password

SECURITY INCIDENT INVESTIGATION REPORT

Multi-Stage Attack Analysis using Wazuh SIEM

- These failed login attempts were successfully captured in Windows Event Logs (Event ID 4625) and correlated in Wazuh, confirming detection of unauthorized access attempts. This activity represents the initial phase of the attack chain.

Category	Details
Who	Unknown threat actor targeting account 'mehrab' on DESKTOP-KA2LJHB
What	Repeated failed authentication attempts — wrong password supplied
When	Intial phase of the attack chain at 2026-04-23 12:44:51 PKT
Where	Local system (loopback interface) - DESKTOP-KA2LJHB
Why	Password guessing / brute force to gain interactive access

Technical Details

Field	Value
Windows Event ID	4625, An account failed to log on
Logon Type	2 (Interactive, Physical console session)
Source IP Address	127.0.0.1 (Loopback - Internal origin)
Failure Status Code	0xC000006D - Wrong username or authentication failure
Sub-Status Code	0xC000006A - Account exists, incorrect password
Caller Process	C:\Windows\System32\svchost.exe (PID 668)
Logon Process	User32
Alert Fire Count	9 times - Indicates automated/scripted activity

Technical Analysis

- Sub-Status 0xC000006A confirms account 'mehrab' EXISTS, attacker has enumerated valid usernames prior to this event.

SECURITY INCIDENT INVESTIGATION REPORT

Multi-Stage Attack Analysis using Wazuh SIEM

- Logon Type 2 (Interactive) originated via loopback (127.0.0.1), attack comes from inside the machine itself.
- 9 consecutive failures within a short window exceeds normal user behavior threshold consistent with automated tooling.
- No variation in target username this is a targeted attack, not random spraying.
- svchost.exe initiating User32 interactive logons is anomalous system services do not normally request console sessions.

Analyst Interpretation

1. The observed pattern is consistent with intentional credential guessing using a script or tool executing locally on the endpoint.
2. The attacker had already identified a valid account name and was systematically attempting to find the correct password.
3. The loopback source address is particularly significant, it indicates the attack is generated from within the compromised or targeted machine, suggesting either an insider threat scenario, a previously implanted malware, or an attacker who already has some level of local code execution.

VERDICT — TRUE POSITIVE

True Positive - Credential Access (Suspicious). Immediate investigation and account lockdown review required.

Recommendations

- Enable Account Lockout Policy via GPO: maximum 3–5 failed attempts before 30-minute lockdown
- Create a Wazuh frequency rule: alert CRITICAL when Event ID 4625 fires 5+ times within 60 seconds for the same account.
- Monitor local authentication anomalies, flag any Event ID 4625 with Source IP 127.0.0.1.

SECURITY INCIDENT INVESTIGATION REPORT

Multi-Stage Attack Analysis using Wazuh SIEM

- Investigate svchost.exe process tree to identify parent process and any associated malicious activity.

Alert 2 — New Account Created | Persistence | Event ID: 4720

- To simulate persistence after gaining access, a new local user account named 'tester' was created on the target system. This step represents a common attacker technique to maintain access even if the original session is terminated.

```
C:\Windows\system32>net user tester myPassword@is123 /add
The password entered is longer than 14 characters. Computers
with Windows prior to Windows 2000 will not be able to use
this account. Do you want to continue this operation? (Y/N) [Y]: y
The command completed successfully.
```

Alert on: Wazuh SEEM

8 hits					
Apr 20, 2026 @ 18:04:33.066 - Apr 27, 2026 @ 18:04:33.066					
Export Formatted	749 columns hidden	Density	1 fields sorted	Full screen	
timestamp	agent.name	rule.description	rule.level	rule.id	
Apr 23, 2026 @ 12:45:29.310	DESKTOP-KA2LJHB	User account enabled or created	8	60109	
Apr 22, 2026 @ 19:59:27.940	DESKTOP-KA2LJHB	User account enabled or created	8	60109	

Document Details		View surrounding documents	View single doc
agent.ip			
agent.name	DESKTOP-KA2LJHB		
data.win.eventdata.accountExpires	%%1794		
data.win.eventdata.displayName	%%1793		
data.win.eventdata.homeDirectory	%%1793		
data.win.eventdata.homePath	%%1793		
data.win.eventdata.logonHours	%%1797		
data.win.eventdata.newUacValue	0x15		
data.win.eventdata.oldUacValue	0x0		
data.win.eventdata.passwordLastSet	%%1794		
data.win.eventdata.primaryGroupId	513		
data.win.eventdata.profilePath	%%1793		
data.win.eventdata.samAccountName	tester		
data.win.eventdata.scriptPath	%%1793		
data.win.eventdata.subjectDomainName	DESKTOP-KA2LJHB		
data.win.eventdata.subjectLogonId	0x8cab0		
data.win.eventdata.subjectUserName	mehrab		
data.win.eventdata.subjectUserSid	S-1-5-21-1410036601-668628055-412612306-1001		
data.win.eventdata.targetDomainName	DESKTOP-KA2LJHB		
data.win.eventdata.targetSid	S-1-5-21-1410036601-668628055-412612306-1011		
data.win.eventdata.targetUserName	tester		

SECURITY INCIDENT INVESTIGATION REPORT

Multi-Stage Attack Analysis using Wazuh SIEM

data.win.system.eventID	4720
data.win.system.eventRecordID	92803
data.win.system.keywords	0x8020000000000000
data.win.system.level	0
data.win.system.message	"A user account was created. Subject: Security ID: S-1-5-21-1410036601-668628055-412612306-1001 Account Name: mehrab Account Domain: DESKTOP-KA2LJHB Logon ID: 0xB0CAB0 New Account: Security ID: S-1-5-21-1410036601-668628055-412612306-1011 Account Name: tester Account Domain: DESKTOP-KA2LJHB Attributes: SAM Account Name: tester Display Name: <value not set> User Principal Name: - Home Directory: <value not set> Home Drive: <value not set> Script Path: <value not set> Profile Path: <value not set> User Workstations: <value not set> Password Last Set: <value not set> Account Expires: <value not set> Primary Group ID: 512 Allowed To
data.win.system.opcode	0
data.win.system.processID	668
data.win.system.providerGuid	{54849625-5478-4994-a5ba-3e3b0328c30d}
data.win.system.providerName	Microsoft-Windows-Security-Auditing
data.win.system.severityValue	AUDIT_SUCCESS
data.win.system.systemTime	2026-04-23T07:45:28.4643480Z
data.win.system.task	13824
data.win.system.threadID	764
data.win.system.version	0
decoder.name	windows_eventchannel
id	1776930329.23751203
input.type	log
location	EventChannel
manager.name	mehrab-virtual-machine
rule.description	User account enabled or created
rule.firedtimes	1
rule.gdpr	IV_35.7.d, IV_32.2
rule.msg13	710

Category	Details
Who	Account 'mehrab' — used to create new account after gaining access
What	New local user account 'tester' created on DESKTOP-KA2LJHB
When	Immediately following the failed logon activity in Alert 1 at 2026-04-23 12:45:29 PKT
Where	On DESKTOP-KA2LJHB
Why	Establishing persistence backup access mechanism or to perform High Privileges Tasks

Technical Details

Field	Value
Windows Event ID	4720, A user account was created
New Account Name	tester

SECURITY INCIDENT INVESTIGATION REPORT

Multi-Stage Attack Analysis using Wazuh SIEM

Field	Value
Password Required	Not required — password policy not enforced
Account Status	Enabled immediately upon creation
Acting Subject	mehrab (authenticated session)

Technical Analysis

- Event ID 4732 → user added to local group
- Group SID S-1-5-32-544 → Administrators (highest privilege)
- Account tester added → same account from previous alert
- Action performed by mehrab → active authenticated session
- Occurred immediately after account creation → sequential attack step
- No approval / admin workflow → direct privilege escalation

Analyst Interpretation

- Confirms intentional privilege escalation, not misconfiguration.
- tester now acts as fully privileged backdoor account.
- Strong correlation with previous alert → single attack chain

VERDICT — CRITICAL

True Positive — Malicious Persistence Activity. Account 'tester' should be immediately disabled and removed.

Recommendations

- Disable and delete account 'tester' immediately , verify no active sessions exist before deletion.
- Enforce strong password policies via GPO minimum complexity, no blank passwords.
- Alert on Event ID 4720 (account creation) in Wazuh, flag any account created outside of approved business hours or change windows.

SECURITY INCIDENT INVESTIGATION REPORT

Multi-Stage Attack Analysis using Wazuh SIEM

- Restrict account provisioning permissions to dedicated IT administrators only.

Alert 3 — Privilege Escalation | Admin Group Addition | Event ID: 4732

- To simulate privilege escalation, the previously created user account 'tester' was added to the local Administrators group. This action represents an attacker elevating privileges to gain full control over the system.

```
C:\Windows\system32>net localgroup administrators tester /add
The command completed successfully.
```

Alert On WAZUH SIEM:

8 hits				
Apr 20, 2026 @ 18:07:01.027 - Apr 27, 2026 @ 18:07:01.027				
Export Formatted	749 columns hidden	Density	1 fields sorted	Full screen
timestamp	agent.name	rule.description	rule.level	rule.id
Apr 23, 2026 @ 12:46:14.353	DESKTOP-KA2LJHB	Administrators Group Changed	12	60154
Apr 23, 2026 @ 12:45:29.362	DESKTOP-KA2LJHB	Users Group Changed	5	60170

Document Details		View surrounding documents	View single document
Name			
data.win.eventdata.subjectLogonId	0x8cab0		
data.win.eventdata.subjectUserName	mehrab		
data.win.eventdata.subjectUserSid	S-1-5-21-1410036601-668628055-412612306-1001		
data.win.eventdata.targetDomainName	Builtin		
data.win.eventdata.targetSid	S-1-5-32-544		
data.win.eventdata.targetUserName	Administrators		
data.win.system.channel	Security		
data.win.system.computer	DESKTOP-KA2LJHB.lab.local		
data.win.system.eventID	4732		
data.win.system.eventRecordID	92869		
data.win.system.keywords	0x8020000000000000		
data.win.system.level	0		
data.win.system.message	"A member was added to a security-enabled local group. Subject: Security ID: S-1-5-21-1410036601-668628055-412612306-1001 Account Name: mehrab Account Domain: DESKTOP-KA2LJHB Logon ID: 0x8CAB0 Member: Security ID: S-1-5-21-1410036601-668628055-412612306-1011 Account Name: - Group: Security ID: S-1-5-32-544 Group Name: Administrators Group Domain: Builtin Additional Information: Privileges: -"		
data.win.system.opcode	0		
data.win.system.processID	668		
data.win.system.providerGuid	{54849625-5478-4994-a5ba-3e3b0328c30d}		
data.win.system.providerName	Microsoft-Windows-Security-Auditing		
data.win.system.severityValue	AUDIT_SUCCESS		
data.win.system.systemTime	2026-04-23T07:46:14.1212194Z		
data.win.system.task	13826		
data.win.system.threadID	2240		
data.win.system.version	0		

SECURITY INCIDENT INVESTIGATION REPORT

Multi-Stage Attack Analysis using Wazuh SIEM

Category	Details
Who	Account 'mehrab' added newly created account to Administrators group
What	User 'tester' added to built-in local Administrators security group
When	After account creation at 2026-04-23 12:46:14 PKT, part of same attack sequence
Where	Windows Local Group Management on DESKTOP-KA2LJHB
Why	Privilege escalation, granting full system control to the backdoor account

Technical Details

Field	Value
Windows Event ID	4732 :A member was added to a security-enabled local group
Target Group	Aministrators (Built-in — SID: S-1-5-32-544)
Account Added	Tester (SID-linked — same account as Alert 2)
Acting Subject	Mehrab, performing the group membership change
Group SID	Confirms built-in local Administrators group

Technical Analysis

- Group SID S-1-5-32-544 unambiguously identifies the built-in Administrators group, this is the highest privilege level on a local Windows system.
- Escalation occurs within seconds of account creation, deliberate and sequential, not accidental.
- Administrator membership grants: full system control, ability to modify security settings, log access, and install software.
- No delay, approval workflow, or justification ticket observed abnormal for any legitimate environment.
- SID linkage directly ties the 'tester' account from Alert 2 to this escalation event — confirms single attack sequence.

SECURITY INCIDENT INVESTIGATION REPORT

Multi-Stage Attack Analysis using Wazuh SIEM

Analyst Interpretation

1. This alert represents the transition from persistence to full control. By adding 'tester' to the Administrators group, the attacker has secured a fully-privileged backdoor account on the system.
2. This is a critical escalation point, from this moment, the attacker has unrestricted access to all system resources, can disable security controls, read all files, and perform any action including the subsequent payload execution and log clearing observed in later alerts.

VERDICT — CRITICAL

CRITICAL — True Positive: Privilege Escalation Confirmed. System must be treated as fully compromised from this point forward.

Recommendations

- Remove 'tester' from Administrators group immediately and disable the account.
- Review all Administrators group members and audit for any other unauthorized additions.
- Monitor Event ID 4732 in Wazuh, create high-priority alerts for any modification to privileged groups.
- Implement Just-in-Time (JIT) access for administrator privileges, no standing admin accounts.
- Apply Principle of Least Privilege (PoLP) across all user accounts in the domain.

Alert 4 — Malware Detection (EICAR) | Execution Phase | Event ID: 1116

- To simulate malware execution in a controlled environment, the standard EICAR test file was downloaded from the internet and executed on the target system. In this scenario, the EICAR test file was intentionally treated as a malicious file to simulate real-world malware detection. Although EICAR is a harmless standardized test string, it is recognized by antivirus solutions as a threat. This

SECURITY INCIDENT INVESTIGATION REPORT

Multi-Stage Attack Analysis using Wazuh SIEM

allows safe validation of detection, alert generation, and SIEM correlation without introducing actual malicious code into the environment.

Alert on WAZUH SIEM

agent.name	rule.description	rule.level	rule.id
DESKTOP-KA2LJHB	Malware detected by Windows Defender (EL...	12	999999
DESKTOP-KA2LJHB	Malware detected by Windows Defender (EL...	12	999999
DESKTOP-KA2LJHB	Malware detected by Windows Defender (EL...	12	999999
DESKTOP-KA2LJHB	Malware detected by Windows Defender (EL...	12	999999
DESKTOP-KA2LJHB	Malware detected by Windows Defender (EL...	12	999999
DESKTOP-KA2LJHB	Malware detected by Windows Defender (EL...	12	999999
DESKTOP-KA2LJHB	Malware detected by Windows Defender (EL...	12	999999
DESKTOP-KA2LJHB	Malware detected by Windows Defender (EL...	12	999999

Document Details		View surrounding documents	View single document
data.win.eventdata.additional Actions ID	0		
data.win.eventdata.additional Actions String	No additional actions required		
data.win.eventdata.category ID	42		
data.win.eventdata.category Name	Virus		
data.win.eventdata.detection ID	{A4B1926F-2F44-4B6C-8E3C-35B67AC10C20}		
data.win.eventdata.detection Time	2025-12-08T06:22:58.689Z		
data.win.eventdata.detection User	DESKTOP-KA2LJHB\mehrab		
data.win.eventdata.engine Version	AM: 1.1.25090.3001, NIS: 1.1.25090.3001		
data.win.eventdata.error Code	0x00000000		
data.win.eventdata.error Description	The operation completed successfully.		
data.win.eventdata.execution ID	1		
data.win.eventdata.execution Name	Suspended		
data.win.eventdata.fwLink	https://go.microsoft.com/fwlink/?linkid=37020&name=Virus:DOS/EICAR_Test_File&threatid=2147519003&enterprise=0		
data.win.eventdata.origin ID	1		
data.win.eventdata.origin Name	Local machine		
data.win.eventdata.path	file_C:\Program Files (x86)\ossec-agent\leicar.com		
data.win.eventdata.post Clean Status	0		
data.win.eventdata.pre Execution Status	3		
data.win.eventdata.process Name	C:\Windows\System32\notepad.exe		
data.win.eventdata.product Name	Microsoft Defender Antivirus		

SECURITY INCIDENT INVESTIGATION REPORT

Multi-Stage Attack Analysis using Wazuh SIEM

Category	Details
Who	Threat actor using elevated 'tester' account context
What	EICAR test file detected and blocked by Windows Defender AV
When	Post-escalation, execution phase of the attack chain
Where	OSSEC agent directory on DESKTOP-KA2LJHB
Why	Payload execution / malware deployment simulation

Technical Details

Field	Value
Windows Event ID	1116 - Windows Defender detected malware
Threat Name	EICAR Test File (Simulated malware payload)
Executing Process	notepad.exe
File Location	OSSEC agent directory (non-standard, suspicious staging)
AV Response	Blocked (real-time protection active)
Detection Source	Windows Defender, real-time scan

Technical Analysis

- Detection triggered by Windows Defender real-time protection, confirms AV pipeline is functional.
- notepad.exe executing a file in the OSSEC agent directory is highly suspicious , process-to-path relationship is abnormal.
- File stored in OSSEC directory is a common attacker technique to abuse trusted directories and evade basic path-based exclusions.
- Execution was blocked, however, the attempt reaching this stage confirms the attacker had successfully reached the execution phase.

SECURITY INCIDENT INVESTIGATION REPORT

Multi-Stage Attack Analysis using Wazuh SIEM

Analyst Interpretation

1. While the EICAR file is benign by design, its use in this simulation validates two critical findings: the attacker reached the payload execution phase, and the AV detection pipeline successfully identified and blocked the threat.
2. In a real attack, the EICAR file would be replaced with functional malware (ransomware, RAT, or data exfiltration tool).
3. The use of notepad.exe as the execution vehicle and the OSSEC directory as a staging location are tradecraft indicators worth monitoring.

VERDICT - TRUE POSITIVE (SIMULATED MALWARE)

The activity represents a successful detection of a simulated malicious payload (EICAR test file).

While the file itself is non-harmful, it was intentionally treated as malware to replicate real-world payload execution. Detection behavior accurately reflects how an actual malicious file would be identified and handled in a production environment.

Recommendations

- Implement Application Control (AppLocker / Windows Defender Application Control) to restrict execution paths.

Alert 5 — Account Deletion | Cleanup / Anti-Forensics | Event ID: 4726

- To simulate account cleanup and removal of access, the previously created user account 'tester' was deleted from the system. This action represents attacker behavior where accounts are removed after use to reduce traces.

```
C:\Windows\system32>net user tester /delete
The command completed successfully.
```

SECURITY INCIDENT INVESTIGATION REPORT

Multi-Stage Attack Analysis using Wazuh SIEM

Alert on Wazuh SIEM:

7 hits					
Apr 27, 2025 @ 18:10:47.536 - Apr 27, 2026 @ 18:10:47.536					
Export Formatted	749 columns hidden	Density	1 fields sorted	Full screen	
timestamp	agent.name	rule.description	rule.level	rule.id	
Apr 23, 2026 @ 12:54:10.520	DESKTOP-KA2LJHB	User account disabled or deleted	8	60111	
Apr 23, 2026 @ 12:35:22.981	DESKTOP-KA2LJHB	User account disabled or deleted	8	60111	
Apr 22, 2026 @ 16:39:12.231	DESKTOP-KA2LJHB	User account disabled or deleted	8	60111	

agent.name	DESKTOP-KA2LJHB
data.win.eventdata.subjectDomain Name	DESKTOP-KA2LJHB
data.win.eventdata.subjectLogonId	0x8cab0
data.win.eventdata.subjectUserName	mehrab
data.win.eventdata.subjectUserSid	S-1-5-21-1410036601-668628055-412612306-1001
data.win.eventdata.targetDomainName	DESKTOP-KA2LJHB
data.win.eventdata.targetSid	S-1-5-21-1410036601-668628055-412612306-1011
data.win.eventdata.targetUserName	tester
data.win.system.channel	Security
data.win.system.computer	DESKTOP-KA2LJHB.lab.local
data.win.system.eventID	4726
data.win.system.eventRecordID	93162
data.win.system.keywords	0x8020000000000000
data.win.system.level	0
data.win.system.message	"A user account was deleted. Subject: Security ID: S-1-5-21-1410036601-668628055-412612306-1001 Account Name: mehrob Account Domain: DESKTOP-KA2LJHB Logon ID: 0x8CAB0 Target Account: Security ID: S-1-5-21-1410036601-668628055-412612306-1011 Account Name: tester Account Domain: DESKTOP-KA2LJHB Additional Information: Privileges -"
data.win.system.opcode	0
data.win.system.processID	668
data.win.system.providerGuid	{54849625-5478-4994-a5ba-3e3b0328c30d}
data.win.system.providerName	Microsoft-Windows-Security-Auditing
data.win.system.severityValue	AUDIT_SUCCESS
data.win.system.systemTime	2026-04-23T07:54:09.5769546Z

Category	Details
Who	Account 'mehrab' deleting the backdoor account after objectives achieved
What	User account 'tester' deleted from the local system
When	After payload execution (cleanup phase), at 2026-04-23 12:54:10 PKT
Where	Windows User Management on DESKTOP-KA2LJHB
Why	Removing evidence of persistence mechanism / anti-forensic cleanup

SECURITY INCIDENT INVESTIGATION REPORT

Multi-Stage Attack Analysis using Wazuh SIEM

Technical Details

Field	Value
Windows Event ID	4726 — A user account was deleted
Deleted Account	tester
Acting Subject	Mehrab(performing the deletion)
Account Lifecycle	Created (4720) → Escalated (4732) → Deleted (4726)

Technical Analysis

- The complete account lifecycle from creation, escalation, and deletion is observed within a a short time.
- Deletion occurs immediately after the execution phase, indicating the attacker achieved their objective and is now cleaning up
- No legitimate administrative workflow involves creating a privileged account, using it briefly, and immediately deleting it
- Account deletion removes the primary persistence artifact but does NOT erase the Windows Security event logs. yet
- This anti-forensic action demonstrates attacker awareness of defensive monitoring and audit procedures

Analyst Interpretation

1. The deletion of account 'tester' is a deliberate anti-forensic action. The attacker, having completed their primary objectives, is now systematically removing evidence of their presence.
2. This behavior pattern is a strong indicator of an experienced threat actor who understands that orphaned accounts are high-visibility IOCs that defenders commonly hunt for.
3. The deletion alone would remove the most visible artifact — however, the event logs remain, which leads directly to the final alert.

SECURITY INCIDENT INVESTIGATION REPORT

Multi-Stage Attack Analysis using Wazuh SIEM

VERDICT — TRUE POSITIVE

True Positive — Malicious Anti-Forensic Activity. Confirms deliberate post-exploitation cleanup behavior.

Recommendations

- Maintain Wazuh log retention for minimum 90 days, deletion of accounts does not erase SIEM records.
- Monitor abnormal account lifecycle patterns, flag any create-privilege-delete sequences.
- Ensure Windows Security logs are forwarded to Wazuh in real-time to prevent log tampering from destroying evidence.

Alert 6 — Security Log Cleared | Defense Evasion | Event ID: 1102

- To simulate defense evasion techniques, the Windows Security event logs were cleared using system commands. This action represents an attempt by an attacker to remove forensic evidence and hinder incident investigation.

```
C:\Windows\system32>wevtutil cl Security
C:\Windows\system32>
```

Alert on Wazuh SIEM:

	Apr 23, 2026 @ 12:54:32.160	DESKTOP-KA2LJHB	The audit log was cleared	5	63103
	Apr 23, 2026 @ 12:36:21.775	DESKTOP-KA2LJHB	The audit log was cleared	5	63103

SECURITY INCIDENT INVESTIGATION REPORT

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data.win.logFileCleared.subjectLogonId	0x8cab0
data.win.logFileCleared.subjectUserName	mehrab
data.win.logFileCleared.subjectUserSid	S-1-5-21-1410036601-668628055-412612306-1001
data.win.system.channel	Security
data.win.system.computer	DESKTOP-KA2LJHB.lab.local
data.win.system.eventID	1102
data.win.system.eventRecordID	93165
data.win.system.keywords	0x4020000000000000
data.win.system.level	4
data.win.system.message	"The audit log was cleared. Subject: Security ID: S-1-5-21-1410036601-668628055-412612306-1001 Account Name: mehrib Domain Name: DESKTOP-KA2LJHB Logon ID: 0x8CAB0"
data.win.system.opcode	0
data.win.system.processID	964
data.win.system.providerGuid	{fc65ddd8-d6ef-4962-83d5-6e5cfe9ce148}
data.win.system.providerName	Microsoft-Windows-Eventlog
data.win.system.severityValue	INFORMATION
data.win.system.systemTime	2026-04-23T07:54:32.268577Z
data.win.system.task	104
data.win.system.threadID	1440
data.win.system.version	1
decoder.name	windows_eventchannel
id	1776930872.24959641
input.type	log
location	EventChannel

Category	Details
Who	Administrator-level account confirmed by privilege requirement to clear logs
What	Windows Security Event Log completely cleared
When	Final stage - immediately after account deletion cleanup,2026-04-23 12:54:32 PKT
Where	Windows Security Event Log on DESKTOP-KA2LJHB
Why	Destroying forensic evidence removing all recorded attack activity from local logs

Technical Details

Field	Value
Windows Event ID	1102 — The audit log was cleared
Log Type Cleared	Windows Security Event Log
Privilege Required	Administrator or SecurityPrivilege - confirms full escalation

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Field	Value
Occurrence	Immediately following account deletion (Alert 5)
SIEM Impact	Local logs destroyed , however Wazuh SIEM retained all forwarded events

Technical Analysis

- Log clearing requires Security Privilege or Administrator membership, this single event confirms that the privilege escalation in Alert 3 was successful.
- Clearing the Security log removes ALL recorded 4625, 4720, 4732, 1116, and 4726 events from the local machine.
- Event ID 1102 is itself generated at the moment of clearing in a local-only scenario, this would be the only remaining trace.
- This action is extremely rare in legitimate operations no authorized business process involves clearing Security event logs.
- Because Wazuh forwards events in real-time, the local log clear has no impact on SIEM records all six alerts are preserved

Analyst Interpretation

1. The clearing of Windows Security event logs is the most decisive indicator of malicious intent in this entire chain. It requires elevated privileges, confirming the success of the escalation chain.
2. It demonstrates the attacker's full awareness of Windows logging mechanisms and their intent to prevent local forensic investigation.
3. This is a hallmark behavior of sophisticated threat actors and is categorized as Defense Evasion under MITRE ATT&CK (T1070.001 — Indicator Removal: Clear Windows Event Logs).

SECURITY INCIDENT INVESTIGATION REPORT

Multi-Stage Attack Analysis using Wazuh SIEM

VERDICT — CRITICAL

CRITICAL — True Positive: Defense Evasion Confirmed. Retroactively confirms full compromise chain.

Recommendations

- Alert immediately on Event ID 1102, this event should trigger a P1/Critical incident response automatically.
- Restrict Security log clearing permissions via GPO remove from all accounts except dedicated audit accounts.
- Consider Wazuh Active Response to auto-isolate the endpoint upon detection of Event ID 1102.

4. ATTACK CHAIN CORRELATION

#	Event ID	Phase	MITRE Tactic	Description & Chain Significance
1	4625	Credential Access	T1110.001	Brute force against account 'mehrab'. Attacker gains foothold. Enables all subsequent steps.
2	4720	Persistence	T1136.001	Backdoor account 'tester' created. Provides persistent access independent of primary account.
3	4732	Privilege Escalation	T1098	'tester' added to Administrators. Attacker now has full system control. Critical turning point.
4	1116	Execution	T1059.001	EICAR payload executed via notepad.exe in OSSEC dir. AV blocks but execution stage reached.
5	4726	Cleanup	T1070	Backdoor account 'tester' deleted. Primary persistence artifact removed. Anti-forensic cleanup begins.
6	1102	Defense Evasion	T1070.001	Security event logs cleared. All local forensic evidence destroyed. Confirms admin-level access.

SECURITY INCIDENT INVESTIGATION REPORT

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Note: MITRE technique IDs are inferred from behavioral patterns. Exact sub-techniques may vary based on tooling used.

5. CORRELATION OBSERVATIONS

- No alert in this chain exists in isolation each event is both the consequence of the previous action and the prerequisite for the next.
- Alert 1 (4625) provides the authenticated session context used in Alerts 2 and 3 without successful credential access, no account creation would be possible.
- Alert 2 (4720) creates the specific account that is modified in Alert 3, the SID linkage between 4720 and 4732 is the definitive proof of intentional sequencing.
- Alert 3 (4732) grants the Administrator privilege that is REQUIRED to perform Alert 6 (1102 log clearing) retroactively confirming the escalation was successful.
- Alert 4 (1116) confirms the system reached the post-exploitation execution phase the attacker had unrestricted system access at this point.
- Alert 5 (4726) and Alert 6 (1102) together represent a complete anti-forensic cleanup sequence designed to eliminate local evidence.
- The temporal compression of all six events into a single session window is the strongest indicator of automated or scripted attack execution.